

ETFs and Futures

Dr. David C. Shimko

FRE 6103
Valuation for Financial Engineers

Trading Places with Eddie Murphy

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Outline

- Equity portfolio trading problems
- ETFs and equity portfolios
- Uses of S&P 500 futures contracts
  - S&P 500 is the widest followed equity index in the U.S.
- Pricing of futures and the GVE
- Implied dividends from equity index futures prices
- Arbitrage pricing bounds
- Simulating risk in futures and forwards

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Some equity strategies…

Strategy	Challenge
An investor holds no stocks currently but wants to speculate on the stock market as a whole rising or falling	The costs of buying or selling all the stocks in the market quickly is prohibitively high, and good execution is unlikely
A well-diversified equity investor owns a lot of stock now, but is concerned about the direction of the market in the short term due to trade wars. They want to own stocks in the long term but don't want to own them now	If they sold now and re-bought the stocks, this would be double the transaction costs!

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SPDR® ETFs (courtesy State Street)

SPDR® Product Lineup

ETF Listing — March 31, 2020

Ticker		Fund Name	Net Expense Ratio (%)
Low-Cost Core			
US Equities			
SPTM	•	SPDR Portfolio S&P 1500® Composite Stock Market	0.03
SPLG	•	SPDR Portfolio S&P 500®	0.03
SPMD		SPDR Portfolio S&P 400® Mid Cap	0.05
SPSM	•	SPDR Portfolio S&P 600® Small Cap	0.05
SPYG	•	SPDR Portfolio S&P 500 Growth	0.04
SPYV	•	SPDR Portfolio S&P 500 Value	0.04
SPYD		SPDR Portfolio S&P 500 High Dividend	0.07
International Equities			
SPGM		SPDR Portfolio MSCI Global Stock Market	0.09
SPDW	•	SPDR Portfolio Developed World ex-US	0.04
SPEU	•	SPDR Portfolio Europe	0.09
SPEM	•	SPDR Portfolio Emerging Markets	0.11
Fixed Income			
SPAB		SPDR Portfolio Aggregate Bond	0.04
SPSB		SPDR Portfolio Short Term Corporate Bond	0.07
SPIB		SPDR Portfolio Intermediate Term Corporate Bond	0.07
SPLB		SPDR Portfolio Long Term Corporate Bond	0.07
SPBO		SPDR Portfolio Corporate Bond	0.06
SPTS		SPDR Portfolio Short Term Treasury	0.06
SPTI		SPDR Portfolio Intermediate Term Treasury	0.06
SPTL		SPDR Portfolio Long Term Treasury	0.06
SPMB	■	SPDR Portfolio Mortgage Backed Bond	0.06
SPHY		SPDR Portfolio High Yield Bond	0.15
SPIP		SPDR Portfolio TIPS	0.12

Ticker		Fund Name	Net Expense Ratio (%)
US Equities			
Core			
SPTM	•	SPDR Portfolio S&P 1500 Composite Stock Market	0.03
SPLG	•	SPDR Portfolio S&P 500	0.03
SPMD		SPDR Portfolio S&P 400 Mid Cap	0.05
SPSM	•	SPDR Portfolio S&P 600 Small Cap	0.05
SPY	•	SPDR S&P 500	0.0945
MDY	•	SPDR S&P MidCap 400®	0.23
SLY	■	SPDR S&P® 600 Small Cap	0.05
DIA	•	SPDR Dow Jones® Industrial Average	0.16
SYE	▲	SPDR MFS Systematic Core Equity	0.60
Style			
SPYG	•	SPDR Portfolio S&P 500 Growth	0.04
SPYV	•	SPDR Portfolio S&P 500 Value	0.04
MDYG		SPDR S&P 400 Mid Cap Growth	0.15
MDYV		SPDR S&P 400 Mid Cap Value	0.15
SLYG		SPDR S&P 600 Small Cap Growth	0.15
SLYV		SPDR S&P 600 Small Cap Value	0.15
SYG	▲	SPDR MFS Systematic Growth Equity	0.60
SYV	▲	SPDR MFS Systematic Value Equity	0.60
Sector			
XLC	•	Communication Services Select Sector SPDR	0.13
XLP	•	Consumer Staples Select Sector SPDR	0.13
XLY	•	Consumer Discretionary Select Sector SPDR	0.13
XLE	•	Energy Select Sector SPDR	0.13
XLF	•	Financial Select Sector SPDR	0.13

"fake corporation"

- An ETF is a *trust* which holds a portfolio of securities
  - Abbr: Exchange Traded Fund
  - Passively managed (or rule-based)
- A *trust* is a simple form of a corporation
- Investors purchase shares in the trust for cash & receive dividends
- Instead of purchasing 500 stocks individually, they can own the entire index with one transaction
- ETFs can also be sold short
- ETFs charge relatively small fees

https://www.ssga.com/library-content/products/fund-docs/etfs/us/information-schedules/spdr-etf-listing.pdf



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Mutual Funds vs ETFs

ETF

- Passively managed
- Able to short (and no uptick ~~tule~~ rule)
- Low fees
- Trade continuously

Mutual Fund

- Most are actively managed
- No ability to short
- Most have higher fees
- Trade only at end of day

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ETFs trade like individual stocks…

SPY SPDR S&P 500 ETF TRUST

$303.30 -8.75 (-2.80%)

Dealer quote -"dealer buys at low price"

Bid x Size **$303.28** x 500

"dealer sells at this price"

Ask x Size **$303.30** x 600

Real-time quote: Jun 24, 2020, 12:09 PM ET

tight bid-ask spread indicates a liquid market

Snapshot | Charts | News | Options | Performance | Tax & Income | Risk & Ratings | Portfolio | Historical Quote

Morningstar on SPY
Last Reported 6/24/20

Fund Report | Prospectus

Open	309.84	Market Cap	266.9 B
Previous Close	312.05	Shares Outstanding	880.0 M
Volume	56,342,662	Yield (dividend yield)	1.83%
Avg Volume (10 days)	121,446,777	SEC 30 Day Yield	1.99%
Day Range	302.10 - 310.51	Quarterly Dividend	1.3662
52 - Week Range	218.26 - 339.08 3/23/20 - 2/19/20	Ex-Dividend Date	6/19/20
		Dividend Payable Date	7/31/20



1D | 5D | 3M | 1Y | 3Y | 5Y

Advanced Chart

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The SPY ETF over time tracks the market



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What are the ETF solutions?

Strategy	Solution
An investor holds no stocks currently but wants to speculate on the stock market as a whole rising or falling	Rising: Go long (buy) SPY ETF's Falling: Go short SPY ETF's * wait * Cover your short (buy back positions)
A well-diversified equity investor owns a lot of stock now, but is concerned about the direction of the market in the short term due to trade wars. They want to own stocks in the long term but don't want to own them now	Don't want to hedge custom portfolio, want to hedge market risk. KEEP MY PORTFOLIO Go short SPY ETFs to protect against overall market drop. * wait Cover your position

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Which is better?

Feature\Product	Stocks	ETFs
Flexibility		
Diversification		
Funding requirement		
Dividends		
Bid-offer spreads		
Commissions		
Ongoing fees		
Transparency		
P/L realization*	On sale or short cover	Same

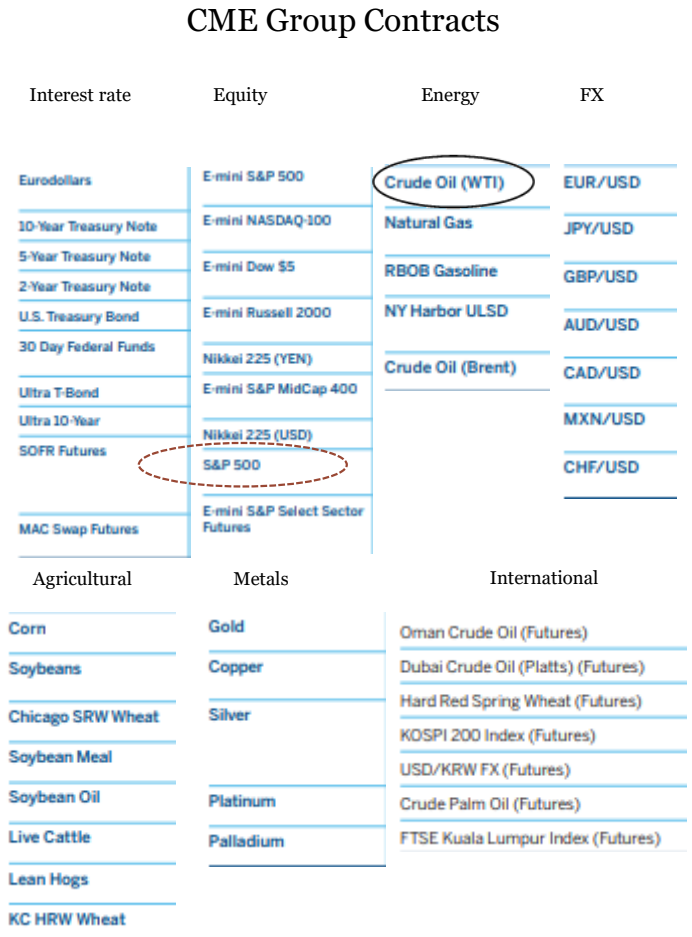
*P/L = profit or loss

Can futures contracts also be used to take equity risk?

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Largest futures & option exchanges in 2026

1. National Stock Exchange of India (NSE Group) — India
2. BSE — India
3. B3 (Brasil, Bolsa, Balcão) — Brazil
4. CME Group — United States
5. Intercontinental Exchange (ICE) — United States / UK
6. Cboe Global Markets — United States
7. Nasdaq — United States / Nordic countries
8. Korea Exchange (KRX) — South Korea
9. Zhengzhou Commodity Exchange (ZCE) — China
10. Shanghai Futures Exchange (SHFE Group) — China
11. Dalian Commodity Exchange (DCE) — China
12. Eurex — Germany
13. Borsa Istanbul — Turkey
14. Miami International Holdings (MIAX) — United States
15. TMX Group — Canada / United States
16. Multi Commodity Exchange of India (MCX) — India
17. Hong Kong Exchanges and Clearing (HKEX) — Hong Kong / UK
18. Japan Exchange Group (JPX) — Japan
19. Taiwan Futures Exchange (TAIFEX) — Taiwan
20. Members Exchange (MEMX) — United States



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What is a futures contract?

A transaction that takes place today or very soon is called a SPOT transaction.

This is a FORWARD transaction.

A FUTURES transaction is an exchange-traded forward transaction.

A standardized and tradable contract… (only difference is the price)
that commits the buyer (going long) and the seller (going short)
to a purchase and sale on a future date
with terms specified today.
esp. price

Example:

Oil futures
1000 barrels
West Texas Intermediate Crude Oil
Physically Delivered in Cushing Oklahoma relatively rare to be delivered
On Dec 21, 2026
At $70 per barrel

If price doubles, I can sell (short) to someone else, profit $70,000!!!

I NEVER HAVE TO TOUCH A DROP OF OIL.

Buy oil:
Toxic, corrosive, explosive, transport, insurance, compliance, storage....
Short oil? hahaha

Example:

S&P E-mini futures
$50 x S&P index value
Quoted in index points, e.g. 6750
No delivery
Cash–settled based on index

Commodity Futures ←----------→ Financial Futures

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Why do people trade S&P futures?

About S&P 500 Futures and Options

S&P 500 futures and options offer a capital-efficient means to manage exposure to the leading large –cap companies of the U.S. stock market. Based on the underlying Standard & Poor's 500 stock index, which is made up of 500 individual stocks representing the market capitalizations of large companies, the S&P 500 Index is a leading indicator of large-cap U.S. equities.

- Speculators
  - Go long if they think the market will rise, or go short when they expect the market to fall (like stock or ETFs)
  - Use very little capital. Only a small fraction of the position value is collected in collateral.
  - Low cost active trading allows traders to liquidate positions quickly and efficiently
- Hedgers
  - Someone who owns stocks might temporarily short futures to hedge their portfolio risk
  - Profits on their stocks are offset by losses on the futures and vice-versa
- Market-makers (traders) Maintain continuous bid and offer quotes to the market
  - Seek to trade frequently, collecting short-term profits, arbitrage benefits and fees

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How do futures work on a daily basis?

- On 10/14/26, the Dec 2026 CME e-mini S&P 500 futures contract traded at 6750
- There are similar *futures* contracts with longer *maturities (quarterly for S&P)*

  maturity dates

- If you take a *long position* in e-mini S&P futures, then you will profit if the S&P rises
  - Your profit is $50 per $1 rise in the S&P 500 futures price
- If you take a *short position*, you will profit if the S&P falls
- The S&P futures price *tracks* the index closely
- To enter a futures position, you must place *collateral* with your broker, but you do not need to pay for the value of the stocks. Minimum collateral is about 2% of position value.

  Exchange has full discretion.

- Your profit/loss is immediately received or paid at the end of every trading day into or from your *margin account*.

  bank account at exchange (clearinghouse)

- When you want to close the trade, you take an *offsetting position* before maturity

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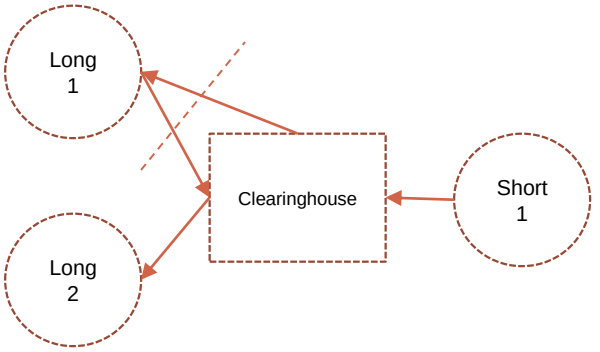
Futures Structure

- Standardized public exchange-traded contract
- Agreement to buy or sell something in the future at a price determined today
  - Some contracts are *cash-settled,* not delivered
- Expected **not** to deliver
  - Only delivers 2-3% of the time
- *Central counterparty* minimizes default risk
  - The central counterparty in futures is called the *clearinghouse*
  - The clearinghouse is the "long to every short and the short to every long"
  - Changes in value of position are immediately received or paid
  - Daily margining increases the security of the clearinghouse

The contract at inception, anonymous to both parties



Long 1 offsets, "books out" or reverses the trade by shorting the same contract prior to expiration



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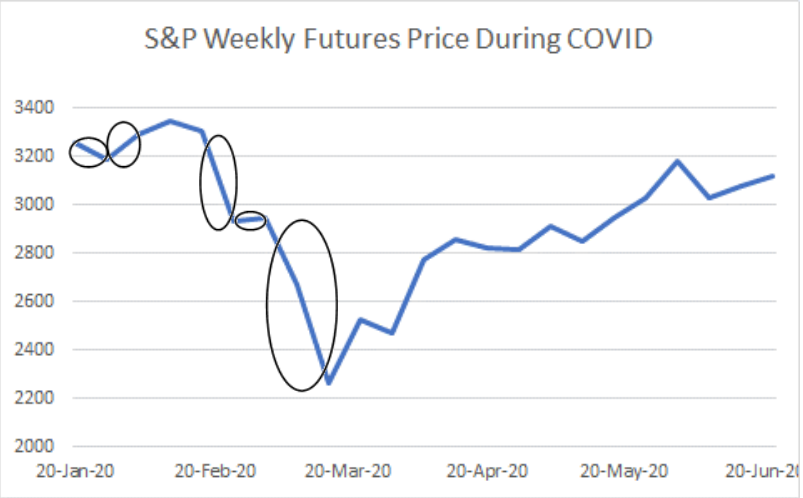
Recap technical terms

Term	Definition
Future/futures	
Maturity	
Long vs buy	
Short vs sell	
Index tracking	
Collateral	
Margin account	
Offsetting position	
(Central) counterparty	
Clearinghouse	
Delivery/Cash settlement	

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Futures simulation

*please note, the 250 multiplier has been discontinued
Minis and micro-minis still trade



Total losses if position closed 6/22/20	33425
Divided by 250	133.7
Initial futures price	3254.2
Closing futures price	3120.5
Change in contract price	-133.7

Total long P/L =
(Closing futs price – Initial futs price) × 250

Margin Account - Long 1 S&P 500 Futures Contract on Jan 20, 2020

Price at inception	3254.2	~$800,000
First day	-69.7	
in Dollars	-17425	

Maintenance margin ignored (2% of posn)

Date	June 2020 Futures	P/L	Account	Min	Deposit
20-Jan-20	3254.2	0	16,271	16,271	16,271
27-Jan-20	3184.5	-17,425	-1,154	15,923	17,077 *
3-Feb-20	3288.1	25,900	41,823	16,441	0
10-Feb-20	3341.6	13,375	55,198	16,708	0
17-Feb-20	3300.8	-10,200	44,997	16,504	0
24-Feb-20	2932.4	-92,100	-47,103	14,662	61,765 *
2-Mar-20	2944.3	2,975	17,637	14,722	0
9-Mar-20	2665.7	-69,650	-52,013	13,329	65,342 *
16-Mar-20	2264.7	-100,250	-86,922	11,324	98,245 *
23-Mar-20	2523.1	64,600	75,924	12,616	0
30-Mar-20	2471	-13,025	62,899	12,355	0
6-Apr-20	2769.8	74,700	137,599	13,849	0
13-Apr-20	2853.8	21,000	158,599	14,269	0
20-Apr-20	2817.3	-9,125	149,474	14,087	0
27-Apr-20	2815.5	-450	149,024	14,078	0
4-May-20	2911.6	24,025	173,049	14,558	0
11-May-20	2850.3	-15,325	157,724	14,252	0
18-May-20	2941.5	22,800	180,524	14,708	0
25-May-20	3029.9	22,100	202,624	15,150	0
1-Jun-20	3179.4	37,375	239,999	15,897	0
8-Jun-20	3028.8	-37,650	202,349	15,144	0
15-Jun-20	3072.9	11,025	213,374	15,365	0
22-Jun-20	3120.5	11,900	225,274	15,603	0

* Margin calls

$259K

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Pricing futures on the S&P 500

First nearby futures
(usu most widely traded, most liquid; aka front month)

Second nearby futures

...and so on

CME Group MARKETS DATA SERVICES EDUCATION

MONTH	OPTIONS	CHARTS	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	HI / LOW LIMIT	UPDATED
SEP 2020	OPT		3106.30	-12.20	3118.50	3126.80	3126.80	3106.30	5	No Limit / 2901.00	08:37:40 CT 24 Jun 2020
DEC 2020	OPT		-	-	3108.60	-	-	-	0	No Limit / 2891.00	08:30:00 CT 24 Jun 2020
MAR 2021	OPT		-	-	3103.00	-	-	-	0	No Limit / 2885.50	08:30:00 CT 24 Jun 2020
JUN 2021	OPT		-	-	3096.20	-	-	-		Limit / 2878.50	08:30:00 CT 24 Jun 2020
SEP 2021	OPT		-	-	3087.60	-					08:30:00 CT Jun 2020
DEC 2021	OPT		-	-	3079.00	-					CT
MAR 2022	OPT		-	-	3070.40						
JUN 2022	OPT		-	-	3061.80						
DEC 2022	OPT		-	-	3044.70	-					CT 2020
DEC 2023	OPT		-	-	3010.30	-					30:00 CT 24 Jun 2020
DEC 2024	OPT		-	-	2975.30	-	-	-	0	No Limit / 2758.00	08:30:00 CT 24 Jun 2020

Legend: OPT Options | Price Chart | About This Report

Important note:

The futures price and the value of a futures contract differ!

Value = 0 at inception
will change over time

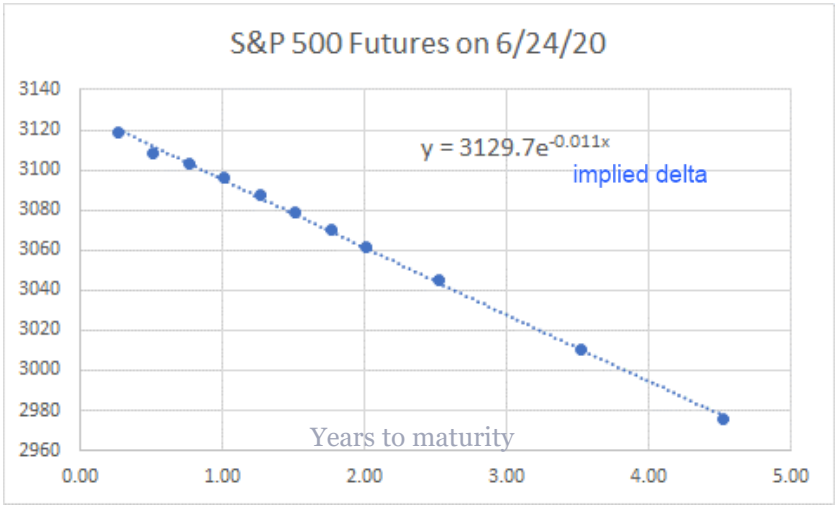
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What are the actual S&P futures prices?

S&P FUTURES by MATURITY DATE

Today	6/24/2020	
	Years	Futures
Sep-20	0.27	3,118.50
Dec-20	0.52	3,108.60
Mar-21	0.77	3,103.00
Jun-21	1.02	3,096.20
Sep-21	1.27	3,087.60
Dec-21	1.52	3,079.00
Mar-22	1.77	3,070.40
Jun-22	2.02	3,061.80
Dec-22	2.52	3,044.70
Dec-23	3.52	3,010.30
Dec-24	4.52	2,975.30

from www.cmegroup.com



Does this mean I can buy the S&P in the future for less than the spot price?

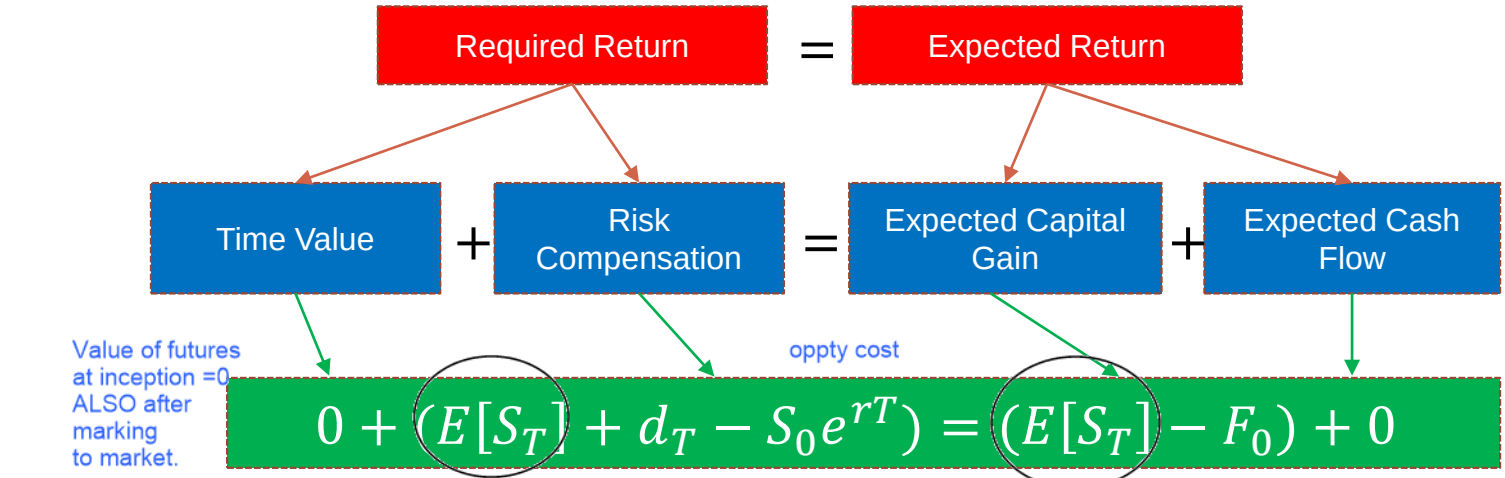
Isn't this a bargain? Don't we expect the market to rise?

How should these contracts be priced?

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Financial forward contract –

Discrete time, continuous compounding, known dividends lost to long position



?

$$F_0 = S_0 e^{rT} - d_T$$

Note that risk neutral pricing obtains because of the risk cancellation in $E[S_T]$

**dollar return notation*

Expected excess $return from stock includes FV dividends

Futures price shows no risk adjustment.

Risk-Neutral Pricing
- Risk affects value (embedded)
- Does not affect formula

Forward Contract Notes

Notation – S_T is the underlying ex-dividend financial asset value at time T

Time value – The value of a forward contract at time 0, with no counterparty risk, $V_0 = 0$

Risk charge – The long forward position takes the same price risk as the one who holds the underlying asset but does not receive dividends

Capital gain – The expected value of the forward position in the future

Cash flow – Assumed dividends are known; d_T is the future value of known dividends, compounded continuously at the risk-free rate. For the continuous case, replace $-dT$ with $*\exp(-\delta T)$

NOTE HOW THE $E[S_T]$ TERMS CANCEL –
EXPECTED FUTURE VALUE INCLUDES RISK COMPENSATION,
BUT ALSO AFFECTS EXPECTED RETURNS

THESE TWO THINGS CANCEL, LEADING TO A RISK-NEUTRAL VALUATION

$$0 + (E[S_T] + d_T - S_0 e^{rT}) = (E[S_T] - F_0) + 0$$

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FINANCIAL FUTURES ONLY

Futures pricing intuition:

Risk transfer with built-in financing

- If I buy stock, the price reflects the risk and expected return of the stock in the future
- With a long futures position, I am committing to take that risk, so there is no incremental risk adjustment

- Since I don't pay cash with futures, this is a benefit causing the futures price to exceed the stock price
- Since I don't receive dividends with futures, this is a deduction from the futures price

- Let r = risk-free interest rate (continuously compounded)
- Let δ = dividend payment rate on stocks (also continuous)

- Then $F(0,T) = S_0 e^{(r-\delta)T}$ (futures price at time 0 for contract maturing at T)

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Implied dividend rate

- The exponential growth rate estimated in the graph is the value of $r - \delta$, or -1.10%
- We can use the long-term treasury yield to estimate r and solve for δ
- Some historical annual dividend yields for the S&P 500 are shown in the table on the right.

Implied $r-\delta = -1.10\%$
5-year treasury bond $r = 0.33\%$
Solve for $\delta = 1.43\%$

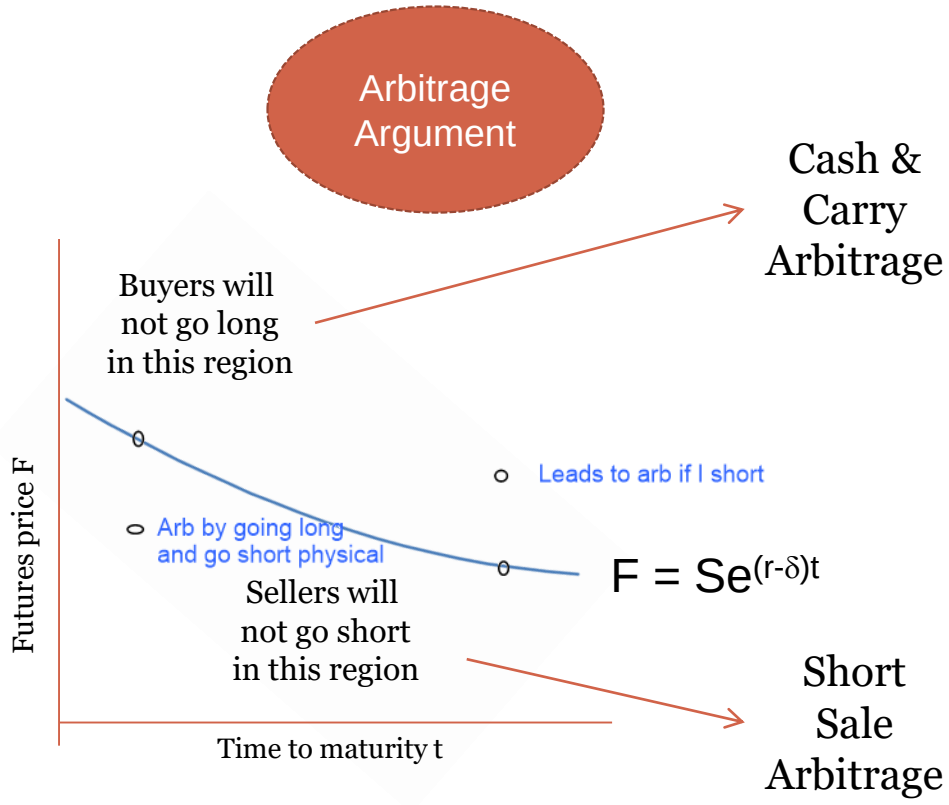
This is the implied dividend yield!

Date	Yield
Jun 25, 2020	1.93% estimate
Dec 31, 2019	1.83%
Dec 31, 2018	2.09%
Dec 31, 2017	1.84%
Dec 31, 2016	2.03%
Dec 31, 2015	2.11%
Dec 31, 2014	1.92%
Dec 31, 2013	1.94%
Dec 31, 2012	2.20%
Dec 31, 2011	2.13%
Dec 31, 2010	1.83%

https://www.multpl.com/s-p-500-dividend-yield/table/by-year

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“No arbitrage” pricing: Market proof



- If $F > Se^{(r-\delta)T}$
  - Borrow money and buy the asset for S
  - Contract to sell forward at F
  - *** Wait to maturity ***
  - Pay interest, get dividends, return money at a cost of of $Se^{(r-\delta)T}$
  - Net arb profit = $F - Se^{(r-\delta)T} > 0$
    - Note there is no market risk
    - This is arbitrage if default is not possible
- If $F < Se^{(r-\delta)T}$
  - Borrow the asset if possible, and sell it
  - Invest cash but pay dividends to owner
  - Contract to buy it back forward at F
  - *** Wait to maturity ***
  - Buy the stock at F and return to owner
  - Net arb profit = $Se^{(r-\delta)T} - F > 0$

Since the left and right sides of the equation cannot be unequal, they must be equal!

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Questions about futures trading

- Do you know your trading counterparty?

  No. (it's the clearinghouse, but you don't know the other transactor)

- What happens when you want to get out of a position?

  If long, go short
  If short, go long

- What if you don't get out of the position before maturity?

  Long - Take delivery.
  Short - Make delivery.

- What if you can't make a margin call?

  Clearinghouse C/H will terminate your position.

  Then close it out and attempt collection. (C/H goes after its own brokers and members if you fail)

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How do ETFs and futures compare?

Feature\Product	S&P Based ETFs	S&P Futures
Flexibility		
Diversification		
S&P index tracking		
Funding requirement		
Dividends		
Bid-offer spreads		
Commissions		
Ongoing fees		
Transparency		
P/L realization		

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Commodity markets are different

- In financial futures markets, arbitrage ensures that prices cannot deviate from their theoretical value, i.e. $F = Se^{(r-\delta)t}$
- Arbitrage requires that we can freely buy and short sell the underlying asset, and store it for long periods of time
- This is easy with stocks, but most commodities violate these assumptions
- For example, it is very expensive to hold oil, storage supply is limited, and it is impossible to short-sell



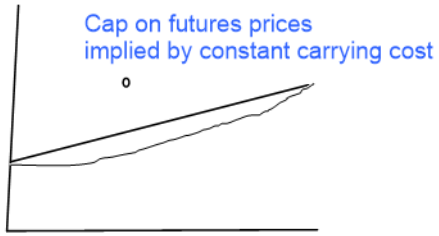
Desk made from two recycled barrels.

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The futures curve for crude oil, 5/13/19

- What determines the upper pricing bound?

For commodities with fixed storage costs, there is a maximum forward price)
For oil, as oil futures rise, the cost of storage rises (in short supply) --- no upper limit

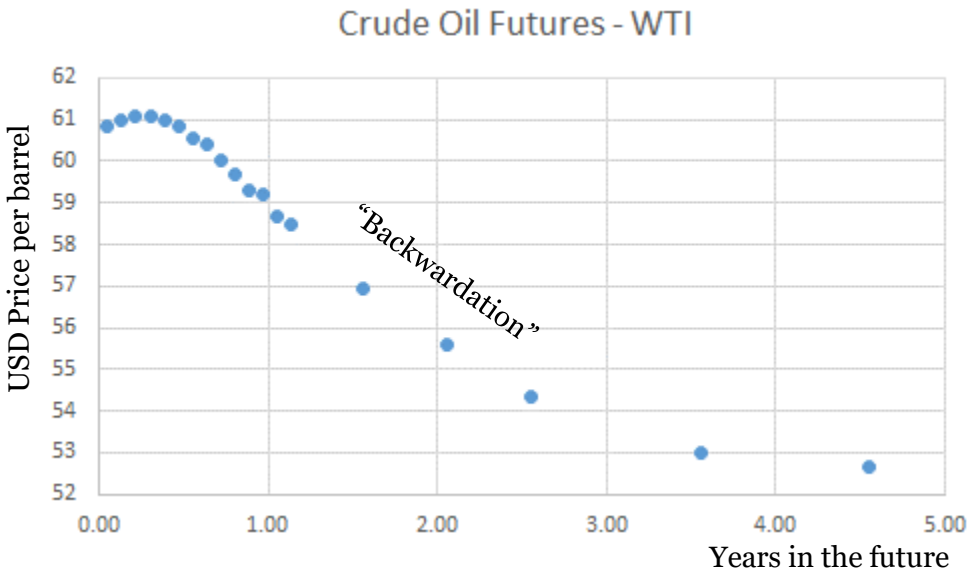


- What determines the lower bound?

Except for gold, I cannot short physical commodities.

No lower limit. No arbitrage bounds.

- Can oil futures prices be negative?



from www.cmegroup.com

- What is the futures price in this (nonfinancial) case?
  - Futures price equals expected future spot price ± risk premium
  - Backwardation could reflect either
    - Market's expectation of falling oil prices
    - Price concession short hedgers are willing to take to reduce risk

Traditional authors used "convenience yield"

My opinion - this puts all the error into a single parameter.

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Super contango

- With no arbitrage bounds active, there is no upper limit on oil futures prices
- High futures prices cause storage costs to rise
  - Note direction of causation!
- Pipelines, trucks and ships are used as storage

Oil prices have moved into 'super contango'

COMMENTARY

John Kilduff | @KilduffReport

Published 10:22 AM ET Wed, 25 Nov 2015 | Updated 1:39 PM ET Wed, 25 Nov 2015

CNBC

It looks to be a volatile final few weeks for crude oil prices. So far, the low for WTI oil prices in 2015 of $37.75 a barrel set in August stands as the low price point -- but not for long.

There is a global supply glut, not just of crude oil, but, increasingly, refined products that will likely break the back of price support in the market, sending oil prices into a holiday plunge. So much so, land based storage tanks are filling up and increasing numbers of volumes are being stored on tankers.

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OIL

‘Scary,’ ‘visceral,’ ‘unprecedented’: Traders describe oil’s wild week and fall to negative prices

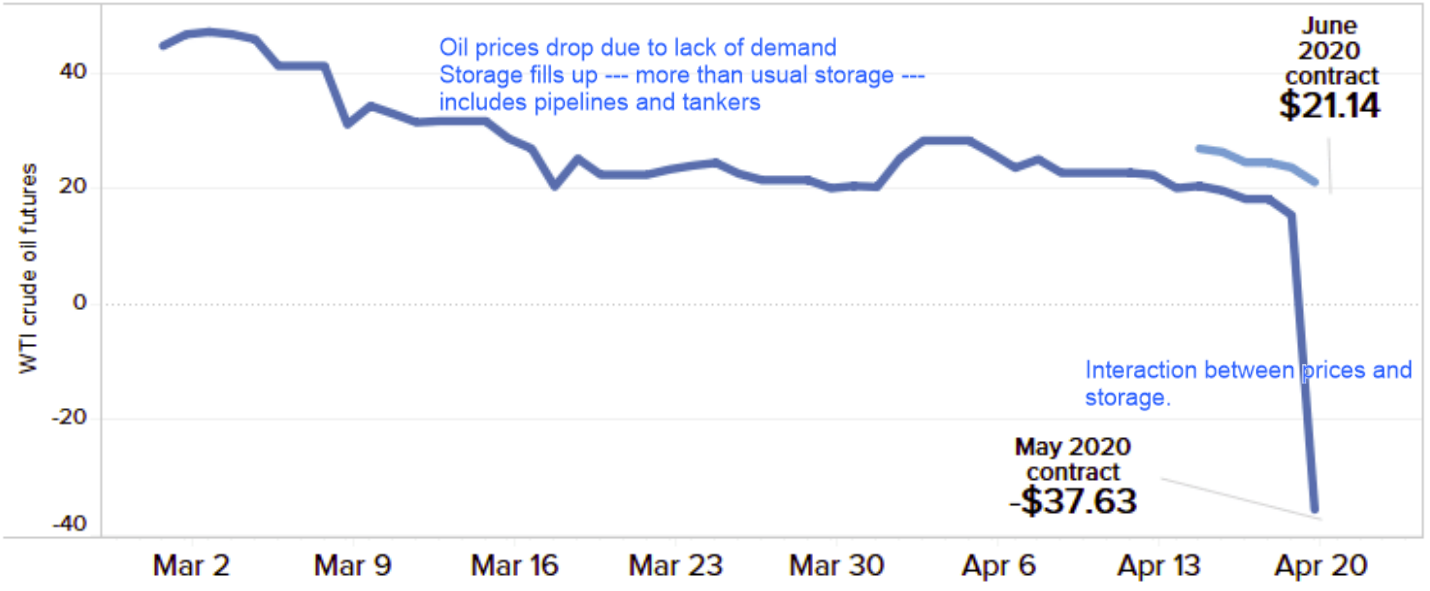
PUBLISHED SAT, APR 25 2020•10:11 AM EDT | UPDATED MON, APR 27 2020•9:17 AM EDT

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SHARE

Oil futures crash

Crude oil WTI futures, dollars per barrel



SOURCE: FactSet, CNBC data

CNBC

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What happened?

- On April 20, 2020, the price of the expiring May West Texas Intermediate (WTI) crude oil futures contract fell below zero for the first time in history, ultimately settling at –$37.63 per barrel. The collapse resulted from an extraordinary combination of factors: the COVID-19 pandemic had sharply reduced global oil demand, oil production remained relatively high, and storage facilities at Cushing, Oklahoma—the delivery point for WTI futures—were approaching capacity. As the contract neared expiration, traders holding long positions who could not accept physical delivery became desperate to liquidate their contracts, effectively paying other market participants to take the oil off their hands.
- The event demonstrated that futures prices can become negative when the costs and constraints associated with physical delivery exceed the commodity's immediate economic value. The consequences included substantial trading losses, severe disruptions to oil markets, increased scrutiny of commodity futures risk management, and changes to trading systems and margin practices to accommodate negative prices. Importantly, the negative settlement was specific to the expiring WTI futures contract, rather than a general indication that all crude oil had become worthless.

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Who trades commodity futures?

- Speculators
  - Think commodity prices are going to rise or fall
  - Use their knowledge of the physical markets to exploit pricing anomalies
- Hedgers
  - Producers short futures on their production to lock in future prices (increasing stability) of cash flows
  - Consumers long futures on their planned purchases to lock in future prices (increasing stability)
  - *Example:* The corn farmer hedges corn prices before the harvest to lock in a profit margin.

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Case Study – Hedging with Corn Futures

- In February, Farmer Brown plans to sow 1,000,000 bu of corn in April, and harvest in July
- Each corn contract is for 5000 bushels
- What position should Farmer Brown take in corn futures to hedge herself against the price risk?
- Notes:
  - Farmer Brown is located in Illinois, and corn markets trade in Illinois (CME) and Missouri (KCBOT)
  - Current futures price is $4 per bushel
  - Farmer Brown's costs are $2,000,000 including everything, and are prepaid.

- Position — long/short
- Contract month — Jul futures
- Number of contracts (Volume): — 200 contracts
- Location: — CME

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Case Study (2): *To hedge or not to hedge?*

- Suppose the actual harvested price is $3, $4 or $5.
- Show Farmer Brown's profit in each case, hedged and unhedged

Recall: 1 mm bushels, sold forward at $4, $2 MM fixed costs.

Price of Corn at harvest	Unhedged revenue (MM)	Unhedged profit	Hedged revenue (MM)	Contract lifetime Mark-to-market	Hedged profit
$3	$3 mm	1	4	1	2
$4	4	2	4	0	2
$5	5	3	4	-1	2

Happy day!

Whether corn rises or falls, I've locked in $2M

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Case Study (3): *My kingdom for rainfall!*

- What if there is a drought, eliminating the crop? (lack of water)
- Because of the drought, corn prices rise $1 to 5 per bu.

Price of Corn at harvest	Unhedged revenue (MM)	Unhedged profit	Hedged revenue (MM)	Contract lifetime Mark-to-market	Hedged profit
~~$3~~					
~~$4~~					
$5	0	-2	-1	-1	-3

- Farmer Brown can buy $3 mm in crop insurance for $0.5 mm.
- Does this help?

She hedged corn she didn't have!!!
VOLUMETRIC RISK

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Case Study (4): *The best day of my life!*

- The day after Farmer Brown put on the trading position, Donald Trump sent out a tweet
- **Corn is the COVID-19 cure!** Prices went to $100 per bushel!!



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Case Study (5): Or was it?

- What was Farmer Brown's margin requirement that day?
  - Assume Farmer Brown has access to $5 million in cash and loans.
- After a week, Trump was fact-checked, and prices returned to normal
- What was the outcome for Farmer Brown?

What was the margin call?

- Short at $4/bu
- Prices go to $100/bu
- 1 mm bushels.

$96 mm

_____?

MARGINING RISK



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Risks of hedging with futures

- Futures are a popular choice for hedgers
  - Believed to be highly competitive markets with low transaction costs
  - Little to no credit risk due to central counterparty
  - Liquid markets mean easy modifications and strategy reversals
- However, there are three major recurring risks for hedgers

Margining risk — Trump tweet

- Risk that you may have to put up cash margin on a losing position, even if it is offset with a noncash gain on your asset

Basis risk — Price in Kansas City vs Chicago

- Hedging something with futures on an asset different than the one in which you have a position

Volumetric risk — Drought

- Risk that the volume of your asset being hedged may change for some reason unexpectedly

In general, it is a good idea to simulate future margin requirements when you first enter a futures position.

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Takeaways

- A futures contract obligates you to buy or sell something in the future at terms specified today.
- In a futures transaction, the market risk of an asset is transferred from the asset owner to the long position. The relative pricing is "risk-neutral", as reflected by the financial futures pricing formula.
- Commodity futures have no such reliable formula, due to constraints on short sales and possibly storage costs that increase with futures prices ("super-contango").
- People use futures to take leveraged market positions (speculators) or to change their risk profile (hedgers).
- For financial assets, forward price $F_{0,T} = S_0 e^{(r-\delta)T}$
  - Assumes constant discount and dividend rates, continuously compounded
  - For commodities, there is no general formula
- Risks in practice
  - Futures contracts were designed to mitigate market risk, but they create liquidity risk for users and credit risk for exchanges.
  - For those who hedge or speculate with futures, the biggest risks are margining, basis risk and volumetric risk.
- Ultimately, futures are called derivatives in the press, but they really approximate an extreme form of leverage. High borrowing can be very dangerous when the value of the invested asset is volatile.

Derivative Debacles - What caused the demise of these companies?

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